

# Exercises for Quantum Information

## Sheet 3 — Quantum Circuits

The gates  $T$  (*Toffoli gate*) and **CNOT** (*controlled negation*) are defined as follows:

$$T: (x, y, z) \mapsto (x, y, z \oplus (x \wedge y)), \quad \text{CNOT}: (x, y) \mapsto (x, y \oplus x),$$

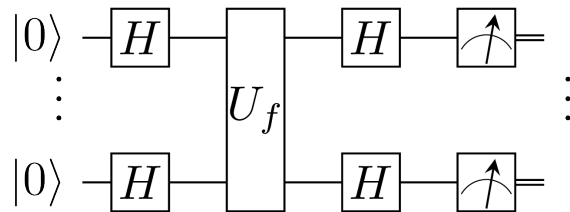
where  $x, y, z$  range over  $\{0, 1\}$ , and  $\oplus$  is a binary XOR operator (*exclusive or*). The **NAND** gate (*not and*) is defined as  $\text{NAND}: (x, y) \mapsto \neg(x \wedge y)$ .

**Exercise 1.** Show that **NAND** is complete, i.e. any boolean function from  $\{0, 1\}^n \rightarrow \{0, 1\}$  can be expressed as a suitable combination of **NAND** gates. Show that the Toffoli gate is reversible (i.e. bijective), and is complete (i.e., for any boolean function  $f: \{0, 1\}^n \rightarrow \{0, 1\}$ , there is a composition of  $T$  gates which takes  $x_0, \dots, x_{n-1}$  with some extra constant bits and outputs  $x_0, \dots, x_{n-1}, f(x_0, \dots, x_{n-1})$  with some extra constant bits).

**Exercise 2.** Compute matrices representing **CNOT** and  $T$  gates. Show that the matrices are unitary.

**Exercise 3.** (Deutsch-Josza problem) Provide a (possibly informal) argument that it is necessary to make at least  $2^{n-1} + 1$  queries when the oracle function  $f$  takes only classical (i.e. 0, 1) inputs. This means that the underlying decision problem is not in P. On the other hand, show that this problem is in coNP. Does this resolve the P versus NP conjecture?

**Exercise 4.** (Bernstein-Vazirani problem, 1992) Given a boolean function  $f: \{0, 1\}^n \rightarrow \{0, 1\}$  which is a dot product between  $\bar{x}$  and some fixed string  $\bar{s}$ , i.e.  $f(\bar{x}) = x_1 s_1 \oplus \dots \oplus x_n s_n$ , compute  $\bar{s}$  by querying  $U_f$  just once. The quantum circuit solving this problem is depicted below (the ancilla qubit is omitted from the picture). Show that measuring the output of this circuit in



the standard basis yields the desired string  $\bar{s}$

**Exercise 5.** Assume you are given a function  $f: \{0, 1\}^n \rightarrow \{0, 1\}$  with a quantum oracle gate  $U_f$  defined as:

$$U_f: (\bar{x}, y) \mapsto (\bar{x}, y \oplus f(\bar{x})).$$

Let  $I_0$  be the set of  $n$ -bit strings with the first bit equal to 0, and  $I_1$  be  $\{0, 1\}^n \setminus I_0$ . You are given a promise that  $f$  fulfills one of the two conditions

1.  $f(\bar{a}) = 0$  if and only if  $\bar{a} \in I_0$ ;
2. the total number of strings from  $I_0$  for which  $f$  is 1 plus the total number of strings from  $I_1$  for which  $f$  is 0 is exactly  $2^{n-1}$ .

Design an algorithm that decides which of the two above conditions is fulfilled via just a single query to  $U_f$ .